

MOHAMMED BOURHYM

Software Engineer

simobourhym01@gmail.com · +212 690 835 989 · github.com/MohammedBourhym
linkedin.com/in/mohammedbourhym · bourhym.pages.dev

Software Engineering graduate from **ENSET Mohammedia** with hands-on experience in **software development**, **AI**, and **cloud computing**. Driven by problem-solving and continuous learning, eager to tackle technical challenges and build reliable solutions that deliver real value.

EDUCATION

Software Engineering — Distributed IT Systems (Cycle d'Ingénieur) — *2023 – 2026*
ENSET Mohammedia, Université Hassan II — *Mohammedia*

DEUG, Computer and Mathematical Science (SMI) — *2021 – 2023*
Faculté des Sciences, Université Abdelmalek Essaâdi — *Tetouan, Morocco*

EXPERIENCES

Software Engineer - PFE Internship — *Feb 2026 - Aug 2026*
Theodo — Technology Consulting & Digital Product Engineering — *Casablanca, Morocco*

- Designed an **AI-powered compliance agent** auditing mobile applications against **App Store** and **Google Play** policies, with a deterministic evaluation framework built on **promptfoo**, automated **GitHub Action** quality gates, and **Python** automation
- Developed a **Next.js/React** authentication system with persistent sessions, automatic **JWT** refresh, **Zod** runtime validation, reusable **Shadcn UI** components, snapshot and automated tests using **Jest** and **React Testing Library**
- Implemented backend features for a **Spring Boot** e-commerce application using **Hexagonal Architecture** and **Domain-Driven Design (DDD)**, including **REST APIs**, **JWT** authentication, and **Liquibase** database migrations. Added unit, integration, and contract tests using **JUnit** and **Testcontainers**, with code quality checks enforced through CI pipelines using **ArchUnit**, **JaCoCo**, and **PIT**

Web Developer — Internship — *Jul 2025 – Sep 2025*
IT Access — IT Services & Software Engineering — *Tangier, Morocco*

- Built a maintenance ticket management application with **Angular** and **Spring Boot** microservices (**Apache Kafka**, **Eureka**, **API Gateway**, **PostgreSQL**), with secure **JWT** authentication and an integrated **Spring AI** assistant (**Google Gemini**)
- Provisioned the full AWS environment with **Terraform** (IaC) — **EKS**, **RDS**, **ECR**, **S3**, **CloudFront**, **VPC** — then containerized services with **Docker**, pushed images to **ECR**, and deployed to **EKS** with **Kubernetes** manifests including health probes, resource limits, and rolling updates
- Automated CI/CD with **GitHub Actions** (build → test → image push → deploy to **EKS**), enforced least-privilege **IAM** permissions, and monitored cluster and application health with **CloudWatch**, **Prometheus**, and **Grafana**

Web Developer — Internship — *Jul 2024 – Aug 2024*
Foliatech — Software Development & SaaS — *Tangier, Morocco*

- Built a full-stack employee management application with **Angular** and **Node.js/Express** — **REST APIs**, cloud-managed **Supabase PostgreSQL/Storage**, **JWT** authentication with refresh token rotation — containerized with **Docker** using multi-stage builds, health checks, and **Docker Compose**
- Deployed on **AWS EC2** behind an **NGINX** reverse proxy with **HTTPS/TLS**, and automated CI/CD with **GitHub Actions** — tests, **Docker** image builds pushed to a registry, and deployment on merge

PROJECTS

AI Public Speaking Coach

- Built a multimodal AI platform for public-speaking analysis, combining **MediaPipe** computer vision (posture, eye contact, gestures), **faster-whisper** speech recognition (transcription, fillers, pacing), and **Google Gemini** for contextual feedback and personalized recommendations
- Architected four cooperating **Flask** services (vision, audio, NLP/LLM, orchestrator) with a **React/TypeScript** 3D-avatar interface, integrating **Google Cloud Text-to-Speech**, synchronized video analysis, and Dockerized deployment

CapLearn — AI Language Learning Application

- Built an AI language-learning application with **React** and **Node.js/Express** — **Groq Whisper** transcription, **FFmpeg** media processing, word-level synchronized subtitles with **SRT** export — and an **MCP Server** exposing **Zod**-validated caption-generation tools for LLM-powered captioning and translation

KNOWLEDGE AND SKILLS

- **Cloud & DevOps:** Cloud & IaC — AWS (EKS, RDS, ECR, S3, CloudFront, VPC), Terraform, Ansible · Containers & Orchestration — Docker, Docker Compose, Kubernetes (EKS), Helm · CI/CD & Version Control — Git, GitHub Actions, Jenkins · Monitoring & Observability — CloudWatch, Prometheus, Grafana
- **OS & Networking:** Linux, Bash Scripting, Networking Fundamentals
- **AI & LLM:** Generative AI, LLM Integration, RAG, AI Agents, Spring AI, LangChain4j, promptfoo
- **Web Development:** HTML, CSS, JavaScript, React, Angular, Spring Boot, Node.js, TailwindCSS
- **Languages & Databases:** Python, Java, C — MySQL, PostgreSQL, MongoDB, Redis, Supabase